Prob. 1 – Trigonometric series

Ex 5 – K.P. Lind

$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,dx \qquad a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)\,dx \qquad b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin(nx)\,dx$

$a_0 = \frac{1}{2L}\int_{-L}^{L} f(x)\,dx \qquad a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos(\frac{n\pi x}{L})\,dx \qquad b_n = \frac{1}{L}\int_{L}^{L} f(x)\sin(\frac{n\pi x}{L})\,dx$

a) $f(x) = \frac{1}{2} + \cos(2x) - 4\sin(4x)$

We can see that $f(x)$ is continuously differentiable $\Rightarrow$ $F(x) = f(x)$

By looking at the function and compairing it with the fourier series we conclude that

$$\frac{1}{2} + \overset{\text{EVEN}}{\cos(2x)} - 4\overset{\text{ODD}}{\sin(4x)} \sim a_0 + \sum_{n=1}^{\infty} a_n \cos(\tfrac{n\pi x}{L}) + b_n \sin(\tfrac{n\pi x}{L})$$

$$T_{\cos(2x)} = \frac{P}{|a|} = \frac{2\pi}{2} = \pi$$

$$T_{\sin(4x)} = \frac{P}{|a|} = \frac{2\pi}{4} = \frac{\pi}{2}$$

$$T_{f(x)} = LCM(\pi, \tfrac{\pi}{2}) = \pi = 2L$$

$$\frac{1}{2} + \cos(2x) - 4\sin(4x) \sim a_0 + \sum_{n=1}^{\infty} a_n \cos(2nx) + b_n \sin(2nx)$$

$$a_0 = \underline{\frac{1}{2}}$$

$$\sum_{n=1}^{\infty} a_n \cos(2nx) = \cos(2x) \Rightarrow a_1 = \underline{1},\; a_n = 0,\; n \neq 1$$

$$\sum_{n=1}^{\infty} b_n \sin(2nx) = -4\sin(4x) \Rightarrow b_2 = \underline{-4},\; b_n = 0,\; n \neq 2$$

b) $f(x) = \overset{\text{EVEN}}{1} + \overset{\text{EVEN}}{\sin^2(x)}$, which has period $T = 2L = \pi \Rightarrow [-\frac{\pi}{2}, \frac{\pi}{2}]$

We can rewrite $f(x)$ as

$$f(x) = 1 + \frac{1-\cos(2x)}{2} = \frac{3}{2} - \frac{\cos(2x)}{2}$$

We do the same as in a)

$$\frac{3}{2} - \frac{\cos(2x)}{2} \sim a_0 + \sum_{n=1}^{\infty} a_n \cos(2nx) + b_n \sin(2nx)$$

$$a_0 = \underline{\frac{3}{2}}$$

$$\sum_{n=1}^{\infty} a_n \cos(2nx) = -\frac{\cos(2x)}{2} \Rightarrow a_2 = \underline{-\frac{1}{2}},\; a_n = 0,\; n \neq 2$$

$$\sum_{n=1}^{\infty} b_n \sin(2nx) = 0 \Rightarrow b_n = \underline{0}$$

c) $$f(x) = \overset{\text{EVEN}}{1} - \overset{\text{ODD}}{x} \quad \text{for} \quad -1 \le x \le 1, \quad \text{with period } T = 2L = 2 \Rightarrow [-1, 1]$$

1 is a constant and x is an odd function. Hence

$a_0 = \underline{1}$

$a_n = \underline{0}$

$$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin(n\pi x)\,dx = \int_{-1}^{1}(1 - x)\sin(n\pi x)\,dx \overset{\int dx = 0}{=} \int_{-1}^{1} -x\sin(n\pi x)\,dx = -2\int_0^1 x\sin(n\pi x)\,dx$$

$$= -2\left[-\frac{x\cos(n\pi x)}{n\pi} + \underbrace{\frac{\sin(n\pi x)}{n^2\pi^2}}_{\Rightarrow 0}\right]_0^1 = \frac{2\cos(n\pi)}{n\pi} = \underline{\frac{2(-1)^n}{n\pi}}$$

d) $$f(x) = \overset{\text{EVEN FUNC.}}{|\cos(3\pi x)|}, \text{ that has period } T = 2L = 1/3 \Rightarrow [-\tfrac{1}{6}, \tfrac{1}{6}]$$

$|\cos(3\pi x)| = \cos(3\pi x)$ on the given interval.

$$a_0 = \frac{1}{2L}\int_{-L}^{L} f(x)\,dx = 3\int_{-\frac{1}{6}}^{\frac{1}{6}}\cos(3\pi x)\,dx = 3\left[\frac{\sin(3\pi x)}{3\pi}\right]_{-\frac{1}{6}}^{\frac{1}{6}} = 3\left(\frac{\sin(3\pi\frac{1}{6})}{3\pi} - \frac{\sin(3\pi(-\frac{1}{6}))}{3\pi}\right) = \underline{\frac{2}{\pi}}$$

$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)dx = 6\int_{-\frac{1}{6}}^{\frac{1}{6}}\cos(3\pi x)\cos(6n\pi x)\,dx = 12\int_0^{\frac{1}{6}}\cos(3\pi x)\cos(6n\pi x)\,dx$$

$2\cos(\alpha)\cos(\beta) = \cos(\alpha+\beta) + \cos(\alpha-\beta)$

$$= \frac{12}{2}\int_0^{\frac{1}{6}}\cos(3\pi x + 6n\pi x) + \cos(3\pi x - 6n\pi x)\,dx = 6\int_0^{\frac{1}{6}}\cos(3\pi x(1+2n)) + \cos(3\pi x(1-2n))\,dx$$

$$= 6\left[\frac{\sin(3\pi x(1+2n))}{3\pi(1+2n)} + \frac{\sin(3\pi x(1+2n))}{3\pi(1-2n)}\right]_{0 \Rightarrow 0}^{\frac{1}{6}} = 6\left(\frac{\sin(3\pi\frac{1}{6}(1+2n))}{3\pi(1+2n)} + \frac{\sin(3\pi\frac{1}{6}(1+2n))}{3\pi(1-2n)}\right)$$

$$= \frac{2\sin(\frac{\pi}{2}+n\pi)}{\pi(1+2n)} + \frac{2\sin(\frac{\pi}{2}-n\pi)}{\pi(1-2n)} = \frac{2}{\pi}\left(\frac{\sin(\frac{\pi}{2}+n\pi)}{1+2n} + \frac{\sin(\frac{\pi}{2}-n\pi)}{1-2n}\right) = \frac{2}{\pi}\left(\frac{(-1)^n}{1+2n} + \frac{(-1)^n}{1-2n}\right) = \frac{2(-1)^n}{\pi}\left(\frac{1}{1+2n} + \frac{1}{1-2n}\right)$$

$$= \frac{2(-1)^n}{\pi}\,\frac{2}{(1-4n^2)} = \underline{\frac{4(-1)^n}{\pi(1-4n^2)}}$$

$b_n = \underline{0}$

Prob. 2 – Complex series

$$c_n = \frac{1}{2L}\int_{-L}^{L} f(x)\,e^{i\frac{n\pi x}{L}}\,dx$$

a) $$f(x) = -3e^{-4ix} + e^{-ix} + 4e^{3ix}$$

We can see that $f(x)$ is continuously differentiable $\Rightarrow F(x) = f(x)$

By looking at the function and comparing it with the Fourier series

we conclude that

$$-3e^{-4ix} + e^{-ix} + 4e^{3ix} \sim \sum c_k e^{i\frac{k\pi}{L}x}$$

$$e^{-4ix} = \cos(4x) - i\sin(4x) \Rightarrow T = \frac{p}{|a|} = \frac{2\pi}{4} = \frac{\pi}{2} = 2L$$

$$e^{-ix} = \cos(x) - i\sin(x) \Rightarrow T = \frac{p}{|a|} = \frac{2\pi}{1} = 2\pi = 2L$$

$$e^{-3ix} = \cos(3x) + i\sin(3x) \Rightarrow T = \frac{P}{|a|} = \frac{2\pi}{3} = \frac{2\pi}{3} = 2L$$

$$T_{f(x)} = 2\pi = 2L$$

$$-3e^{-4ix} + e^{-ix} + 4e^{3ix} \sim \sum c_k e^{ik\pi x}$$

$$c_{-4} = \underline{-3}$$

$$c_1 = \underline{1}$$

$$c_3 = \underline{4}$$

b) $f(x) = \cos^2(x)$, for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$, with per d 1 -

By trig. identities we can rewrite $f(x)$

$$f(x) = \frac{1+\cos(2x)}{2} = \frac{1}{2}\left(1 + \frac{e^{2ix} e^{-2ix}}{2}\right) = \frac{1}{2} + \frac{e^{2ix}}{4} + \frac{e^{-2ix}}{4}$$

By looking at the function and compairing it with the fourier series

we conclude that

$$\frac{1}{2} + \frac{e^{2ix}}{4} + \frac{e^{-2xi}}{4} \sim \sum c_k e^{i\frac{k n\pi}{L}x} = \sum c_k e^{2inx}, \quad L = \frac{\pi}{2}$$

$$c_0 = \underline{\frac{1}{2}}$$

$$\frac{e^{\pm 2ix}}{4} = c_k e^{2inx} \Rightarrow c_{\pm 1} = \underline{\frac{1}{4}}$$

c) $f(x) = \underset{\text{EVEN}}{x^2} - \underset{\text{ODD}}{x}$ for $-1 \leq x \leq 1$, with period $T = 2L = 2$

$$c_n = \frac{1}{2L}\int_{-L}^{L} f(x)e^{\frac{in\pi x}{L}}\,dx = \frac{1}{2}\int_{-1}^{1}(x^2 - x)e^{in\pi x}\,dx = \frac{1}{2}\int_{-1}^{1} x^2 e^{in\pi x}\,dx - \frac{1}{2}\int_{-1}^{1} x e^{in\pi x}\,dx$$

$$= \frac{1}{2}\int_{-1}^{1} x^2 e^{in\pi x}\,dx \qquad \int u'v = uv - \int uv'$$

$$u' = e^{in\pi x} \qquad v = x$$

$$-\frac{1}{in\pi}\int x e^{in\pi x} \qquad u = \frac{e^{in\pi x}}{in\pi} \qquad v' = 1$$

$$\Downarrow$$

$$\left[\frac{x^2 e^{in\pi x}}{2in\pi} + \frac{x e^{in\pi x}}{2 i n\pi^2} - \frac{e^{in\pi x}}{2 n\pi^2}\right]_{-1}^{1}$$

d) $f(x) = e^{-x}$ for $-1 \le x \le 1$, with period $T = 2L = 2$

$$c_n = \frac{1}{2L}\int_{-L}^{L} f(x)\, e^{\frac{in\pi x}{L}}\, dx = \frac{1}{2}\int_{-1}^{1} e^{-x}\cdot e^{in\pi x}\, dx = \frac{1}{2}\int_{-1}^{1} e^{x(in\pi-1)}\, dx = \frac{1}{2}\left[\frac{e^{x(in\pi-1)}}{in\pi-1}\right]_{-1}^{1}$$

$$= \frac{1}{2}\left(\frac{e^{in\pi-1}}{in\pi-1} - \frac{e^{1+in\pi}}{in\pi-1}\right) = \frac{e^{in\pi-1}+e^{1+in\pi}}{2in\pi-2} = \sinh(\qquad \frac{\sinh(1)}{1+in\pi}(-1)^n$$

Prob. 3 — Parseval's theorem

a) Parsevals identity gives us the error

$$E_N = \int_{-L}^{L}(f(x)-F_N(x))^2 dx = \int_{-L}^{L} f^2(x)\,dx - L\left(2a_0^2 + \sum_{n=1}^{N} a_n^2 + b_n^2\right) = \int_{-L}^{L} f^2(x)\,dx - 2L\sum_{n=-N}^{N}|c_n|^2$$

b) $f(x) = e^{-x}$ for $-1 \le x \le 1$, with period $T = 2L = 2$

We calculated the c_n in 2d) to be $c_n = \frac{\sinh(1)}{1+in\pi}(-1)^n$ can remove this because of abs value

$$E_N = \int_{-1}^{1} f^2(x)\,dx - 2\sum_{n=-N}^{N} c_n^2$$

$$\int_{-1}^{1} f^2(x)\,dx = \int_{-1}^{1} e^{-2x}\,dx = \left[-\frac{e^{-2x}}{2}\right]_{-1}^{1} = \frac{-e^{-2}-(-e^{-(-2)})}{2} = \frac{e^{2}-e^{-2}}{2} = \sinh(2) = 3.63$$

$$E_N = 3.63 - 2\sum_{n=-N}^{N} c_n^2 = 3.63 - 2\sum_{n=-N}^{N}\frac{\sinh^2(1)}{(1+in\pi)^2} = 3.63 - \sum\frac{2\sinh^2(1)}{1+n^2\pi^2}$$

$$E_2 = 3.63 - 2(c_{\pm1}^2 + c_{\pm1}^2) = \underline{0.22}$$

$$E_4 = 3.63 - 2(c_1^2 + c_2^2 + c_3^2 + c_4^2) = \underline{0.123}$$

$$E_8 = 3.63 - 2(c_1^2 + c_2^2 + c_3^2 + c_4^2 + c_5^2 + c_6^2 + c_7^2 + c_8^2) = \underline{0.066}$$

c) $f(x) = e^{-|x|}$, $-1 \le x \le 1$, with period $T = 2L = 2$

$$c_n = \frac{1}{2L}\int_{-L}^{L} f(x)\, e^{\frac{in\pi x}{L}}\, dx = \frac{1}{2}\int_{-1}^{1} e^{-|x|}\cdot e^{-in\pi x}\, dx = \frac{1}{2}\int_{0}^{1} e^{-x} e^{-in\pi x}\, dx + \frac{1}{2}\int_{-1}^{0} e^{x} e^{-in\pi x}\, dx$$

$$= \frac{1}{2}\int_{0}^{1} e^{-x(1+in\pi)}\, dx + \frac{1}{2}\int_{-1}^{0} e^{x(1-in\pi)}\, dx = \frac{1}{2}\left[-\frac{e^{-x(1+in\pi)}}{1+in\pi}\right]_{0}^{1} + \frac{1}{2}\left[\frac{e^{x(1-in\pi)}}{1-in\pi}\right]_{0}^{-1} = \frac{1}{2}\left(\frac{1-e^{-(1-in\pi)}}{1-in\pi} + \frac{1-e^{-(1+in\pi)}}{1+in\pi}\right)$$

$$= \frac{1}{1+n^2\pi^2}\left(1-e^{-1}(-1)^n\right)$$

$$\int_{-1}^{1} g^2(x)\,dx = \int_{-1}^{1} e^{2\cdot(-|x|)}\,dx = \int_{-1}^{0} e^{-2x}\,dx + \int_{0}^{1} e^{2x}\,dx = \left[-\frac{e^{-2x}}{2}\right]_{-1}^{0} + \left[\frac{e^{2x}}{2}\right]_{0}^{1} = \underline{e^{2}-1}$$

$$E_N = e^{-2}-1-2\sum_{n=-N}^{N} c_n^2 = e^{-2}-1-2\sum_{n=-N}^{N}\left(\frac{1-e^{-1}(-1)^n}{1+n^2\pi^2}\right)^2$$

d) $g(x)$ is smoother $\Rightarrow$ better approximation

Prob. 4 — Gibbs phenomenon

$$f(x) = 1 - x^2, \quad x \in [0,1] \qquad , L = 1$$
$$= \text{sign}(x)(1-x^2), \quad x \in [-1,0]$$

a) By information provided in the text, we can say that the function is odd. Hence

$$a_0 = 0$$

$$a_n = 0$$

$$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin(n\pi x)\,dx = \frac{2}{L}\int_0^L f(x)\sin(n\pi x)\,dx = 2\int_0^1 (1-x^2)\sin(n\pi x)\,dx$$

$$= 2\int_0^1 \sin(n\pi x)\,dx - 2\int_0^1 x^2\sin(n\pi x)\,dx = 2\left[\frac{\cos(n\pi x)}{n\pi}\right]_0^1 - 2$$

$$\left(\begin{array}{l} \int_0^1 x^2 \sin(n\pi x)\,dx = -\frac{x^2\cos(n\pi x)}{n\pi} - \frac{2}{n\pi}\int_0^1 x\cos(n\pi x)\,dx \\ \int_0^1 x\cos(n\pi x) = \frac{x\sin(n\pi x)}{n\pi} - \frac{2}{n\pi}\int_0^1 \sin(n\pi x) = \frac{2x\sin(n\pi x)}{n^2\pi^2} - \frac{4\cos(n\pi x)}{n^3\pi^3} \end{array}\right)$$

$$b_n = 2\left[\frac{\cos(n\pi x)}{n\pi}\right]_0^1 - 2\left[-\frac{x^2\cos(n\pi x)}{n\pi} + \frac{2x\sin(n\pi x)}{n^2\pi^2} - \frac{4\cos(n\pi x)}{n^3\pi^3}\right]_0^1$$

$$= \frac{2}{n\pi}(\cos(n\pi) - \cancel{1}) - \frac{2}{n\pi}\left(-\cos(n\pi) + \frac{\sin(n\pi)}{n\pi} - \right.$$

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